

Akash Gogate

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EDUCATION

University of Wisconsin at Madison

Graduating: May 2027

Bachelor of Science in Computer Science and Biology • Dean's List Fall 2024, Spring 2025, Fall 2025 • 3.9 GPA

Relevant Coursework: Artificial Intelligence, Deep Learning, Advanced Algorithms, Linear Algebra, Differential Equations, Data Structures, Cryptography

South Brunswick High School

Relevant Coursework: Computer Science Capstone, AP Statistics, AP Biology, Android Application Development, Virtual Reality, Game Development, Web Development

TECHNOLOGIES

Languages: Python, R, Java, JavaScript, Rust, C++, SQL, Linux/Bash

ML & Statistics: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Hugging Face, NLP, Random Forest, KNN, Biostatistics, Multi-Omics Data Integration, Statistical Modeling, Cross-Validation, Statistical Significance, Benchmarking

Bioinformatics: Scanpy, Seurat, CARD, Bioconductor, DESeq2, scVI, scANVI, scGen, AmortizedLDA, scRNA-seq, NGS, Spatial Transcriptomics, Computational Biology, Gene Expression, Differential Expression, Pathway Analysis, Pipeline Development, Reproducible Research, Pipeline Validation

Agentic & LLM: LLM Agents, LLM APIs, RAG, Semantic Similarity Search, Claude Code, AgentBench, Jupyter Notebook, FastAPI, GitHub Actions

RESEARCH EXPERIENCE

The Kendzierski Lab at the University of Wisconsin-Madison

September 2025 - Present

Student Research Intern

- Built hypothesis-driven agentic pipeline for scRNA-seq (NGS) and spatial transcriptomics gene expression analysis across scVI, scANVI, scGen, and AmortizedLDA; systematic pipeline validation and GPU-accelerated profiling cut GPU consumption by **300%**
- Engineered TransferAgent module for an interdisciplinary computational biology paper grading framework, automating citation mapping, peer-reviewed literature synthesis, and cross-paper validity scoring, reducing review time by **~60%** via a reproducible evaluation methodology

Inspirit AI Research Internship

Sep 2022 - Mar 2023

Machine Learning Research Intern

- Built skin cancer detection pipeline using OpenCV preprocessing and feature extraction fed into scikit-learn classifiers (Decision Tree, KNN) achieving **93% classification accuracy** via hyperparameter tuning and cross-validation; published as peer-reviewed paper through Inspirit AI
- Separately trained a Random Forest classifier on miRNA sequences achieving **95% predictive accuracy ($p < 0.05$)** for cancer cell likelihood prediction, validated via stratified k-fold cross-validation across multiple data splits

PUBLICATIONS

"Early Skin Cancer Detection Improvement," Inspirit AI Research Program, 2023

TECHNICAL PROJECTS

Self-Improving LLM Agent (<https://github.com/AkashGogate/SelfImprovingLLMAgent>)

Jan 2026 - May 2026

- Developed novel repeated_failure_rate metric to measure LLM self-improvement efficacy across a **5,000+ task benchmark**; designed LLM-as-judge evaluation harness scoring 7 quality dimensions; failure-driven loop outperformed ReAct baseline by **15pp**

Lotus Health (<https://github.com/AkashGogate/lotus-health>)

March 2026

- Financial risk modeling engine on **1,080-node ICD-10 comorbidity graph** integrating MEPS and CMS insurance data to compute 5-year disease cost paths across **45M records**
- Implemented RAG pipeline using semantic similarity search over ICD-10 graph (SQL-queried NoSQL store) as vector database; object-oriented Python NLP layer separates inference from deterministic logic, eliminating hallucinations in outputs

miRcore Cancer Detection Research

Jul 2021

- Trained a Random Forest classifier in R on micro-RNA expression sequences achieving **95% predictive accuracy ($p < 0.05$)** for cancer cell likelihood prediction, validated via stratified k-fold cross-validation across multiple data splits

INDUSTRY EXPERIENCE

Leidos

May 2025 - Present

Software Engineer Intern

- Optimized satellite observation scheduling via memoization + DP in Python, **reducing constraint violations** while maintaining mission objective alignment under strict real-time operational constraints
- Designed Kafka event-driven pipeline persisting to MongoDB with **fault-tolerant, low-latency streaming**; deployed containerized test suite on Kubernetes via **GitHub Actions CI/CD** with full unit coverage across all components

LEADERSHIP

Princeton Racket Club

May 2024 - Aug 2024

Tennis Coach and Tournament Director

- Directed tournament logistics and scheduling; coached athletes of all skill levels with customized training programs; managed end-to-end client communications and conflict resolution throughout the season

Tennis Racket Stringing Services

Jan 2019 - Present

- Founded and scaled to **50+ clients**; manage all operations, inventory, and client consultations